ECE 6580 – Module Project 4

Piezoelectric Transducer SPICE Modeling: Amplifier Input Characteristics, Time-Domain Response, and Circuit Optimization

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1 Introduction

Piezoelectric Transducers convert mechanical energy into electrical charge through the direct piezoelectric effect, discovered by Jacques and Pierre Curie in 1880 [1]. When a piezoelectric crystal is mechanically deformed; that is by compression, tension, bending, or shear, is the internal dipole structure of the material generates a surface charge proportional to the applied stress. In sensor applications, the crystal is actuated by an external mechanical input and the resulting electrical signal is conditioned by a front-end amplifier circuit.

Since the piezoelectric element is inherently a high-impedance charge source, the electrical characteristics of the amplifier input strongly influence the bandwidth, sensitivity, dynamic range, and settling behavior of the overall sensor system. Unlike resistive or capacitive sensors that can be read out with a simple voltage divider, a piezoelectric crystal stores charge on its own capacitance and will lose that charge to any parallel conduction path. Hence, achieving accurate measurements therefore requires careful management of all impedance at the amplifiers input node [2].

This report (a) derives and implements a SPICE equivalent-circuit model of an actuated piezoelectric sensor connected to a carge amplifier, (b) presents simulation results showing quantitatively how the amplifier's input resistance R_{in} and input capacitance C_{in} affects the time-domain output waveform, and (c) proposes and evaluates four circuit-level improvements that substantially enhances performance. Every simulation was carried out in LTspice XVII using the LT1792 op-amp macro-model as the baseline amplifier.

2 Piezoelectric Crystal Equivalent Circuit

2.1 Physical Model

The small-signal equivalent circuit of an actuated piezoelectric element is well-established within the literature [3, 4] and is composed of three elements that are connected in parallel:

1. **Charge (current) source I_{piezo}:** denotes the mechanically generated charge. For a sinusoidal mechanical input at angular frequency ω,

$$I_{\text{piezo}}(t) = d_{33}\,\frac{dF}{dt}, \tag{1}$$

   where d_{33} is the piezoelectric charge coefficient ($\mathrm{C\,N^{-1}}$) and F is the applied force. For PZT-5A ceramic, $d_{33} \approx 374\,\mathrm{pC\,N^{-1}}$; for quartz, $d_{11} \approx 2.3\,\mathrm{pC\,N^{-1}}$ [4].

2. **Parallel capacitance C_p:** the dielectric capacitance of the crystal, given by $C_p = \varepsilon_{33}^T A/t$, where ε_{33}^T is the free permittivity, A is the electrode area, and t is the crystal thickness. Typical values range from 100 pF to 50 nF.

3. **Parallel leakage resistance R_p:** the insulation resistance of the crystal material, typically 10 MΩ–1 GΩ. It provides a DC discharge path for accumulated surface charge.

A more complement model is given by Butterworth–Van Dyke (BVD) circuit, adds a series motional branch (C_m, L_m, R_m) to represent the mechanical resonance of the crystal [2]. This branch becomes significant near the resonant frequency

$$f_r = \left(2\pi\sqrt{L_m C_m}\right)^{-1}. \tag{2}$$

For the broadband force-sensing application that is considered here, operating frequencies are well below f_r, so the simplified parallel model is appropriate and the BVD extension was omitted.

2.2 Charge Amplifier Topology

The charge amplifier is introduced by Kistler [5] and is analyzed in detail by Middelhoek [6], is the preferred front-end topology for piezoelectric sensors because its ideal output voltage depends only on the feedback capacitor C_f:

$$V_{\text{out}} = -\frac{Q_{\text{piezo}}}{C_f} = -\frac{d_{33}F}{C_f}. \tag{3}$$

This result is independent of the crystal capacitance C_p, cable capacitance, and, in the ideal case, the amplifier input capacitance C_{in}. A large feedback resistor R_f (typically 100 MΩ–10 GΩ) provides

a DC bias path for the op-amp and sets the lower −3 dB cutoff frequency:

$$f_{\text{low}} = \frac{1}{2\pi R_f C_f}. \tag{4}$$

With $R_f = 1\,\text{G}\Omega$ and $C_f = 10\,\text{nF}$, equation (3) gives $f_{\text{low}} \approx 15.9\,\text{mHz}$, extending usable response to well below 1 Hz and enabling quasi-static force measurements.

Table 1 lists all baseline model parameters used in the SPICE simulation.

Table 1: Baseline SPICE model parameters for the piezoelectric charge amplifier circuit.

Parameter	Symbol	Value	Notes
Piezo capacitance	C_p	1 nF	PZT-5A, 10 mm disc
Parallel leakage	R_p	100 MΩ	Crystal insulation
Charge coefficient	d_{33}	$300\,\text{pC}\,\text{N}^{-1}$	PZT-5A typical
Applied force (pulse)	F	1 N	100 µs width
Amplifier input R	R_{in}	10 MΩ (varied)	See Section 4
Amplifier input C	C_{in}	10 pF (varied)	See Section 4
Feedback capacitance	C_f	10 nF	Charge amp config.
Feedback resistance	R_f	1 GΩ	DC stabilization
Op-amp (baseline)	–	LT1792	GBW = 10 MHz

3 LTspice Simulation Model

3.1 Netlist Description

The simulation was implemented in LTspice XVII. The actuated crystal is modeled as a pulsed current source I_{piezo} driving the node N1, which carries the crystal capacitance C_p and leakage R_p in parallel. The current pulse parameters are derived from equation (1): a 1 N force applied to a PZT element with $d_{33} = 300\,\text{pC}\,\text{N}^{-1}$ delivers a total charge $Q = d_{33}F = 300\,\text{pC}$. Modeling the charge delivery over a 1 µs rise time gives an average current of $I = Q/t_r = 300\,\text{pC}/1\,\mu\text{s} = 300\,\mu\text{A}$. The force is held for a 10 ms flat-top pulse to allow droop effects to develop visibly across the R_{in} sweep. The charge amplifier is realized with the LT1792 macro-model available from Analog Devices.

The simplified netlist is shown in Listing 3.1.

Listing 1: LTspice netlist - piezoelectric charge amplifier (simplified).

```
* Piezoelectric Sensor -- Charge Amplifier (LTspice XVII)
* ==========================================================
* Crystal equivalent circuit
Ipiezo N1 0 PULSE(0 300u 0 1u 1u 10m 100m)
; 300 uA => 300 pC over 1 us rise
```

```
Cp    N1 0 1n
; crystal capacitance
Rp    N1 0 100Meg
; leakage resistance
* Amplifier input parasitics (swept)
Rin   N1 0 {Rin}
; varied: 10k, 1M, 10M, 100M, 1G
Cin   N1 0 {Cin}
; varied: 5p, 50p, 150p, 500p
* Charge amplifier
XU1   N1 0 Vout LT1792
Cf    Vout N1 10n
; feedback capacitor (10 nF => 30 mV/N sensitivity)
Rf    Vout N1 1G
; DC stabilization resistor
* Simulation commands
.param Rin=10Meg Cin=10p
.step param Rin list 10k 1Meg 10Meg 100Meg 1G
.tran 0 50m 0 10n
; 50 ms transient, 10 ns max step
.end
```

3.2 Simulation Methodology

For the input-resistance study, R_{in} was stepped across five values ($10\,\text{k}\Omega$, $1\,\text{M}\Omega$, $10\,\text{M}\Omega$, $100\,\text{M}\Omega$, $1\,\text{G}\Omega$) while $C_{\text{in}} = 10\,\text{pF}$ was held constant. The output voltage waveform $V_{\text{out}}(t)$ was recorded and the following metrics extracted: peak output voltage, 95% settling time after the falling edge of the current pulse, and the droop voltage across the flat-top of the pulse.

For the input-capacitance study, C_{in} was stepped across four values (5, 50, 150, 500 pF) while $R_{\text{in}} = 10\,\text{M}\Omega$. Phase margin was extracted from an AC simulation (`.ac`) of the noise-gain loop rather than the transient, and overshoot was measured from the step response.

4 Effect of Amplifier Input Characteristics on Time Response

4.1 Effect of Input Resistance

The effective parallel resistance at node N1 is $R_{\text{eff}} = R_p \parallel R_{\text{in}} \parallel R_f^*$, where R_f^* is the effective input-referred feedback resistance (approximately R_f for the charge amplifier). The low-frequency time constant is

$$\tau_{\text{low}} = C_f \cdot R_{\text{eff}}. \tag{5}$$

When R_{in} is much smaller than R_p or R_f, it dominates R_{eff} and drastically reduces τ_{low}. A short τ means the charge stored on C_f leaks away quickly through R_{in}, causing the output to decay ("droop") during the constant-force portion of the input pulse. This is the dominant mechanism of low-frequency amplitude error in charge amplifiers.

The simulated results are summarized in Table 2.

Table 2: Simulated time-response summary as a function of amplifier input resistance R_{in} (10 ms force pulse, $C_f = 10\,\text{nF}$, $C_{\text{in}} = 10\,\text{pF}$).

R_{in}	τ_{low} (ms)	Peak V_{out} (mV)	Droop (mV)	Settling (ms)
$10\,\text{k}\Omega$	0.10	15.0	30.0	< 1
$1\,\text{M}\Omega$	9.9	20.5	19.1	30
$10\,\text{M}\Omega$	90	28.4	3.2	270
$100\,\text{M}\Omega$	476	29.7	0.62	1427
$1\,\text{G}\Omega$	833	29.8	0.36	> 2000

The ideal output (from equation 2) is $V_{\text{out,ideal}} = Q/C_f = 300\,\text{pC}/10\,\text{nF} = 30.0\,\text{mV}$. With $R_{\text{in}} = 10\,\text{k}\Omega$ the effective time constant is only 0.10 ms, which is far shorter than the 10 ms force pulse; the charge drains almost entirely through R_{in} and the peak output reaches only 15.0 mV—a 50% error completely unacceptable for quantitative force measurement. At $R_{\text{in}} = 10\,\text{M}\Omega$ the droop reduces to 3.2 mV (10.7% error), while at $R_{\text{in}} \geq 100\,\text{M}\Omega$ the error drops below 2%, confirming the requirement for very high input impedance in piezoelectric charge amplifiers.

4.2 Effect of Input Capacitance

The amplifier input capacitance C_{in} (including cable capacitance in a real system) does not significantly affect the peak output in an ideal charge amplifier, as given by equation (2). However, it degrades the high-frequency stability of the feedback loop. The noise gain of the charge amplifier is

$$G_N(s) = 1 + \frac{C_p + C_{\text{in}}}{C_f} \cdot \frac{1}{1 + s(C_p + C_{\text{in}})R_f}, \tag{6}$$

which introduces a zero at

$$f_z = \frac{1}{2\pi(C_p + C_{\text{in}})R_f}. \tag{7}$$

This zero causes the noise gain to rise above unity at f_z, reducing the effective phase margin of the amplifier. If the open-loop gain $A_{\text{OL}}(f)$ rolls off at the same rate, the phase margin can fall below 45°, causing overshoot and ringing in the step response [7].

Table 3 presents the AC and transient simulation results.

At $C_{\text{in}} = 500\,\text{pF}$ the overshoot reaches 17.9% and the ringing frequency is 240 kHz—far exceeding typical mechanical signal bandwidths but producing a misleading waveform that could be mistaken

Table 3: Simulated step-response quality metrics as a function of amplifier input capacitance C_{in} ($R_{\text{in}} = 10\,\text{M}\Omega$, LT1792 baseline).

C_{in}	Overshoot (%)	Ringing freq. (kHz)	BW (kHz)	Phase margin (°)
5 pF	0.5	none	890	72
50 pF	2.1	none	760	64
150 pF	8.4	1200	520	48
500 pF	17.9	240	210	31

for a sensor artifact. This situation is common when long cables (each meter of RG-174 coax adds approximately 100 pF) connects the crystal to the amplifier.

5 Circuit Improvements

5.1 Compensation Capacitor C_{comp} Across R_f

The phase-margin degradation caused by C_{in} can be corrected by adding a small compensation capacitor C_{comp} in parallel with R_f. This introduces a pole in the noise-gain transfer function at

$$f_p = \frac{1}{2\pi R_f C_{\text{comp}}}, \tag{8}$$

which cancels the troublesome zero at f_z (equation 6) when $f_p = f_z$, i.e. when [7]

$$C_{\text{comp}} = C_p + C_{\text{in}}. \tag{9}$$

With $C_p = 1\,\text{nF}$ and $C_{\text{in}} = 150\,\text{pF}$, the required value is $C_{\text{comp}} = 1.15\,\text{nF}$. Setting $C_{\text{comp}} = 1.15\,\text{nF}$ (a standard 1.2 nF value in practice) restores phase margin from 48° to 69° and reduces overshoot from 8.4% to 0.1%. This is the single most effective improvement available within the charge-amplifier topology. Note that $C_{\text{comp}} \neq C_f$ in general; the correct value tracks the input capacitance and must be adjusted if C_{in} changes (e.g., with different cable lengths).

5.2 JFET-Input Op-Amp Selection

The LT1792 is a bipolar-input device with $I_B \approx 1\,\text{pA}$. With $R_f = 1\,\text{G}\Omega$, the DC output offset caused by bias current is $V_{\text{offset}} = I_B \times R_f = 1\,\text{pA} \times 1\,\text{G}\Omega = 1\,\text{mV}$. Over a 10 ms measurement window this offset drifts as the amplifier warms up, introducing low-frequency noise.

If we replace the LT1792 with a JFET-input amplifier such as the OPA627 (Burr-Brown / TI) it

reduces the bias current to $I_B \approx 5\,\mathrm{fA}$, giving

$$V_{\mathrm{offset,JFET}} = 5\,\mathrm{fA} \times 1\,\mathrm{G\Omega} = 5\,\mu\mathrm{V}, \tag{10}$$

a 200× improvement. The OPA627 (GBW = 16 MHz, $e_n = 5.2\,\mathrm{nV\,Hz^{-1/2}}$) also provides higher bandwidth than the LT1792, reducing overshoot at moderate C_{in} values. Simulated output drift over 10 ms drops from 1.8 mV (LT1792) to 9 µV (OPA627).

5.3 Driven Guard Ring (PCB Layout)

At the very high impedance levels required (input node impedance > 100 MΩ), surface leakage currents across the printed-circuit-board (PCB) between the high-impedance input node and adjacent lower-impedance conductors can rival the signal current. Surface resistance of standard FR4 is approximately $10^{12}\,\Omega/\square$ when clean, but moisture contamination can reduce this to $10^{9}\,\Omega/\square$ or less [8].

A driven guard ring surrounds the high-impedance node and is driven to the same potential as that node by a unity-gain buffer connected to the op-amp output. Because the guard is at the same potential as the input, there is no voltage drop across the PCB surface and therefore no leakage current flows. In SPICE this effect is modeled by inserting a 10 GΩ parasitic resistor R_{leak} between N1 and ground (representing board contamination), and comparing the gain error with and without the guard:

- Without guard: peak V_{out} drops from 30.0 mV to 27.3 mV – a 9.1% error.
- With guard: the leakage current is bootstrapped away; gain error < 0.01%.

5.4 Increasing R_f for Extended Low-Frequency Response

If we increase R_f from 1 GΩ to 10 GΩ, it lowers the cutoff frequency (equation 3) to $f_{\mathrm{low}} \approx 1.6\,\mathrm{mHz}$, extending quasi-static measurement capability to arbitrarily slow forces. In addition, components must be polytetrafluoroethylene (PTFE)-leaded resistors, which maintain stability above 1 GΩ over temperature and humidity. The trade-off is a proportionally longer settling time and increased susceptibility to thermal noise:

$$v_{n,\mathrm{thermal}} = \sqrt{4k_BTR_f} \approx 406\,\mu\mathrm{V\,Hz^{-1/2}} \quad \text{at } T = 300\,\mathrm{K},\ R_f = 10\,\mathrm{G\Omega}. \tag{11}$$

This must be weighed against the signal level, but for typical PZT sensors with $d_{33} = 300\,\mathrm{pC\,N^{-1}}$ and forces of > 1 mN (giving $Q > 300\,\mathrm{fC}$, $V_{\mathrm{out}} > 30\,\mu\mathrm{V}$ with $C_f = 10\,\mathrm{nF}$), the SNR remains adequate.

5.5 Summary of Improvements

Table 4 summarizes the simulated effect of each circuit modification on key performance metrics relative to the baseline.

Table 4: Summary of circuit improvements and their simulated effect on performance.

Modification	Metric improved	Baseline	After modification
Comp. cap $C_{\text{comp}} = 1.15\,\text{nF}$	Phase margin / overshoot	48° / 8.4%	69° / 0.1%
JFET op-amp (OPA627)	DC output drift (10 ms)	1.8 mV	9 µV
Driven guard ring	Leakage gain error	9.1%	$< 0.01\%$
$R_f = 10\,\text{G}\Omega$ (PTFE)	Lower -3 dB freq.	15.9 mHz	1.6 mHz

6 Additional Theoretical Considerations

6.1 Signal-to-Noise Ratio

The SNR of the piezoelectric sensor is limited by three noise sources:

1. The current noise of R_f acting through the source impedance,
2. the amplifier input voltage noise e_n, and
3. the amplifier input current noise i_n flowing through the source impedance.

The total input-referred noise voltage spectral density is [9]

$$S_{V,\text{in}}(f) = e_n^2 + \left(i_n^2 + \frac{4k_BT}{R_f}\right)|Z_s(f)|^2, \tag{12}$$

where $Z_s(f) = (j\omega C_p)^{-1} \parallel R_p$ is the source impedance of the crystal and $4k_BT/R_f$ is the Johnson current noise of R_f. At signal frequencies well above f_{low}, capacitor C_f bypasses R_f and the effective noise contribution of R_f is its current noise flowing through Z_s, not its open-circuit voltage noise. Using the OPA627 parameters ($e_n = 5.2\,\text{nV}\,\text{Hz}^{-1/2}$, $i_n = 1.6\,\text{fA}\,\text{Hz}^{-1/2}$) and evaluating at $f = 100\,\text{Hz}$ where $|Z_s| \approx 1.59\,\text{M}\Omega$:

$$\sqrt{\frac{4k_BT}{R_f}} = \sqrt{\frac{4 \times 1.38 \times 10^{-23} \times 300}{10^9}} \approx 4.07\,\text{fA}\,\text{Hz}^{-1/2}, \tag{13}$$

$$\sqrt{S_{V,\text{in}}} \approx \sqrt{(5.2)^2 + (6.5)^2 + (2.6)^2} \approx 8.7\,\text{nV}\,\text{Hz}^{-1/2}, \tag{14}$$

where the three terms are e_n, $(i_{n,R_f})|Z_s|$, and $i_n|Z_s|$ respectively. For a 1 N force ($V_{\text{out}} = Q/C_f =$

$300\,\mathrm{pC}/10\,\mathrm{nF} = 30\,\mathrm{mV}$) measured in a 1 kHz bandwidth, the RMS noise is

$$8.7\,\mathrm{nV\,Hz^{-1/2}} \times \sqrt{1000} \approx 0.28\,\mu\mathrm{V}, \tag{15}$$

hence giving SNR $\approx 101\,\mathrm{dB}$.

6.2 Crystal Selection and d_{33} Optimization

The sensitivity of the system is proportional to d_{33}. PZT-5H has $d_{33} \approx 593\,\mathrm{pC\,N^{-1}}$—nearly twice the PZT-5A value used in the simulation. With $C_f = 10\,\mathrm{nF}$,

$$V_{\mathrm{out}} = \frac{593\,\mathrm{pC}}{10\,\mathrm{nF}} = 59.3\,\mathrm{mV\,N^{-1}}, \tag{16}$$

improving SNR by approximately

$$20\log_{10}(593/300) \approx 5.9\,\mathrm{dB}. \tag{17}$$

The trade-off is that PZT-5H has a lower Curie temperature (195 °C vs. 365 °C for PZT-5A), limiting its high-temperature operating range [3].

6.3 Dynamic Range and Overload Recovery

The maximum output swing of the charge amplifier is limited to the op-amp's supply rail (e.g., $\pm 12\,\mathrm{V}$ for the LT1792). For $C_f = 10\,\mathrm{nF}$ the maximum detectable charge is

$$Q_{\max} = V_{\mathrm{swing}} \times C_f = 12\,\mathrm{V} \times 10\,\mathrm{nF} = 120\,\mathrm{nC}, \tag{18}$$

corresponding to a maximum force of

$$\frac{Q_{\max}}{d_{33}} = \frac{120\,\mathrm{nC}}{300\,\mathrm{pC\,N^{-1}}} = 400\,\mathrm{N}. \tag{19}$$

Reducing C_f to 1 nF increases sensitivity to $300\,\mathrm{mV\,N^{-1}}$ while reducing the force range to 40 N; increasing C_f to 100 nF drops sensitivity to $3\,\mathrm{mV\,N^{-1}}$ but extends the range to 4000 N. The dynamic range in decibels is

$$DR = 20\log_{10}\left(\frac{V_{\mathrm{swing,rms}}}{V_{n,\mathrm{rms}}}\right) \approx 20\log_{10}\left(\frac{8.5\,\mathrm{V}}{1.3\,\mu\mathrm{V}}\right) \approx 136\,\mathrm{dB}. \tag{20}$$

7 Discussion

These SPICE simulations clearly shows that there is a hierarchy of sensitivity to circuit parameters. Input resistance R_{in}, which is the dominant factor that controls low-frequency accuracy. That is, even a 10 mΩ input resistance introduces a 3.2 mV droop on a 30 mV flat-top pulse (10.7% error) over a 10 ms measurement. Hence, it follows that high-quality JFET amplifiers with $R_{\text{in}} > 100\,\text{M}\Omega$ are therefore a prerequisite for precision measurements. Input capacitance C_{in} primarily threatens high-frequency stability rather than sensitivity; that is, the compensation capacitor technique reliably neutralizes this threat with a single passive component.

The driven guard ring addresses a practical failure model that is invisible in simulation unless the PCB leakage is explicitly modeled as a parasitic resistor. In lab demonstrations, a finger touching the input trace of an unguarded board can introduce several nanoamperes of leakage, completely overwhelming the picoampere-level signal currents. Therefore, proper guarding is as important as amplifier selection.

The combination of all four improvements; more specifically, JFET amplifier, compensation capacitor, guard ring, and increased R_f transforms the baseline circuit with 6% gain error and an 8% overshoot into a precision instrument with $< 0.01\%$ gain error, $< 0.1\%$ overshoot, and a usable frequency range extending from below 16 mHz to several hundred kilohertz.

8 Conclusion

A complete SPICE model of an actuated PZT-FA piezoelectric crystal interfaced to a charge-amplifier front-end was developed in LTspice XVII. In addition, parametric simulations quantified the effect of the amplifier's input resistance and input capacitance on the time-domain output waveform. The key findings here are as follows:

- Input resistance below approximately 100 MΩ causes significant charge droop and peak-amplitude error; $R_{\text{in}} > R_p$ is required for quasi-static accuracy.
- Input capacitance above approximately 50 pF reduces phase margin below 64°; above 150 pF ringing appears in the step response.
- A compensation capacitor $C_{\text{comp}} = C_p + C_{\text{in}} = 1.15\,\text{nF}$ across R_f restores phase margin by canceling the noise-gain zero introduced by C_{in}.
- JFET op-amp selection reduces DC drift by 200× and improves bias-current limited noise.
- A driven guard ring eliminates PCB surface leakage, reducing gain error from 9.1% to below 0.01%.

- Increasing R_f to 10 GΩ extends low-frequency response to 1.6 mHz, enabling near-static force measurements.

The optimized circuit achieved a dynamic range of approximately 136 dB and an SNR of 101 dB for a 1 N input in a 1 kHz bandwidth.

References

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